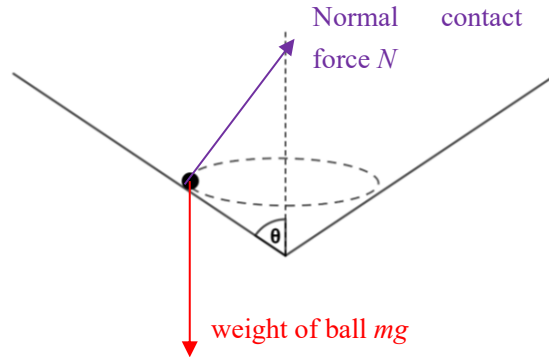


2(b)(i)



Horizontal component of normal contact force provides the **centripetal force**:

$$N \cos \theta = m r \omega^2 \quad \text{--- (1)}$$

The **weight is balanced** by the **vertical component of the normal contact force**:

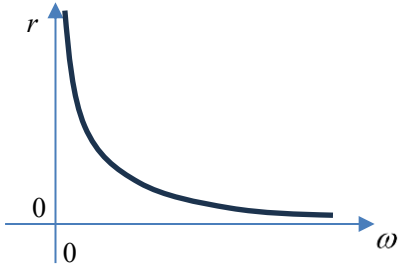
$$N \sin \theta = m g \quad \text{--- (2)}$$

$$(2) / (1) : \quad \tan \theta = \frac{g}{r \omega^2}$$

B1

C1

A0

CKW MC	- Many candidates did not provide proper statements for the “show” question. - Quite a number of candidates recognized that $\tan \theta$ is a ratio of mg and $mr\omega^2$ mathematically. However, they do not go on to explain how $\tan \theta$ comes about from Newton’s 2nd law, in the horizontal direction, and Newton’s 1st law, in the vertical direction.	
2(b)(ii)	$\omega = 2\pi f = 2\pi(3) = 6\pi \text{ rad s}^{-1}$ Since $v = r\omega$, $\tan \theta = \frac{g}{r\omega^2} = \frac{9.81}{0.10(6\pi)^2}$ $\theta = 15^\circ$	C1 A1
CKW MC	This part was generally well done, except for careless mistakes in calculations, e.g. forgetting to square, or conversion of units.	
2(c)		A1
CKW MC	$\tan \theta = \frac{g}{r\omega^2} \rightarrow$ For same θ and g, $r \propto \frac{1}{\omega^2}$. The graphs shows that as ω increases r decreases. - Some candidates did not realise there were 2 asymptotes. - Many candidates did not label the origin.	